MEASURES OF DISPERSION

Learning Outcome

- calculate, interpret, and evaluate measures of dispersion to address an investment problem

Few would disagree with the importance of expected return or mean return in investments: To understand an investment more completely, however, we also need to know how returns are dispersed around the mean. **Dispersion** is the variability around the central tendency. If mean return addresses reward, then dispersion addresses risk and uncertainty.

In this lesson, we examine the most common measures of dispersion: range, mean absolute deviation, variance, and standard deviation. These are all measures of **absolute dispersion**. Absolute dispersion is the amount of variability present without comparison to any reference point or benchmark.

The Range

We encountered range earlier when we discussed the construction of frequency distributions. It is the simplest of all the measures of dispersion.

Definition of Range. *The **range** is the difference between the maximum and minimum values in a dataset:*

$$\text{Range} = \text{Maximum value} - \text{Minimum value}. \tag{2}$$

An alternative way to report the range is to specify both the maximum and minimum values. This alternative definition provides more information as the range is reported as "from Maximum Value to Minimum Value."

One advantage of the range is ease of computation. A disadvantage is that the range uses only two pieces of information from the distribution. It cannot tell us how the data are distributed (i.e., the shape of the distribution). Because the range is the difference between the maximum and minimum values in the dataset, it is also sensitive to extremely large or small observations ("outliers") that may not be representative of the distribution.

Mean Absolute Deviations

Measures of dispersion can be computed using all the observations in the distribution rather than just the highest and lowest. We could compute measures of dispersion as the arithmetic average of the deviations around the mean, but the problem is that deviations around the mean always sum to 0. Therefore, we need to find a way to address the problem of negative deviations canceling out positive deviations.

One solution is to examine the absolute deviations around the mean as in the **mean absolute deviation** (MAD).

MAD Formula. *The MAD for a sample is:*

$$\text{MAD} = \frac{\sum_{i=1}^{n} |X_i - \overline{X}|}{n}, \tag{3}$$

where $\overline{X}$ is the sample mean, n is the number of observations in the sample, and the | | indicate the absolute value of what is contained within these bars.

The MAD uses all of the observations in the sample and is thus superior to the range as a measure of dispersion. One technical drawback of MAD is that it is difficult to manipulate mathematically compared with the next measure we will introduce, sample variance.

Sample Variance and Sample Standard Deviation

A second approach to the problem of positive and negative deviations canceling out is to square them. Variance and standard deviation, which are based on squared deviations, are the two most widely used measures of dispersion. **Variance** is defined as the average of the squared deviations around the mean. **Standard deviation** is the square root of the variance.

Sample Variance

In investments, we often do not know the mean of a population of interest, so we estimate it using the mean from a sample drawn from the population. The corresponding measure of dispersion is the sample variance or standard deviation.

Sample Variance Formula*. The **sample variance**, s^2, is:*

$$s^2 = \frac{\sum_{i=1}^{n} (X_i - \overline{X})^2}{n-1}, \tag{4}$$

where $\overline{X}$ is the sample mean and n is the number of observations in the sample.

statistical properties of the sample variance. The quantity $n - 1$ is also known as the number of degrees of freedom in estimating the population variance. To estimate the population variance with s^2, we must first calculate the sample mean, which itself is an estimated parameter. Therefore, once we have computed the sample mean, there are only $n - 1$ independent pieces of information from the sample; that is, if you know the sample mean and $n - 1$ of the observations, you could calculate the missing sample observation.

Sample Standard Deviation

Variance is measured in squared units associated with the mean, and we need a way to return to those original units. Standard deviation, the square root of the variance, solves this problem and is more easily interpreted than the variance.

A useful property of the sample standard deviation is that, unlike sample variance, it is expressed in the same unit as the data itself. If the dataset is percentage of daily returns for an index, then both the average and the standard deviation of the dataset is in percentage terms, while the variance is in squared percentage of daily returns.

Sample Standard Deviation Formula. *The* ***sample standard deviation****, s, is:*

$$s = \sqrt{\frac{\sum_{i=1}^{n} (X_i - \bar{X})^2}{n-1}}, \qquad (5)$$

where $\bar{X}$ is the sample mean and n is the number of observations in the sample.

Because the standard deviation is a measure of dispersion about the arithmetic mean, we usually present the arithmetic mean and standard deviation together when summarizing data. When we are dealing with data that represent a time series of percentage changes, presenting the geometric mean (i.e., representing the compound rate of growth) is also very helpful.

Downside Deviation and Coefficient of Variation

An asset's variance or standard deviation of returns is often interpreted as a measure of the asset's risk. Variance and standard deviation of returns take account of returns above and below the mean, or upside and downside risks, respectively. However, investors are typically concerned only with **downside risk**—for example, returns below the mean or below some specified minimum target return. As a result, analysts have developed measures of downside risk.

Downside Deviation

In practice, we may be concerned with values of return (or another variable) below some level other than the mean. For example, if our return objective is 6.0 percent annually (our minimum acceptable return), then we may be concerned particularly with returns below 6.0 percent a year. The target downside deviation, also referred to as the **target semideviation**, is a measure of dispersion of the observations (here, returns) below a target—for example 6.0 percent. To calculate a sample target semideviation, we first specify the target. After identifying observations below the target, we find the sum of those squared negative deviations from the target, divide that sum by the total number of observations in the sample minus 1, and, finally, take the square root.

Sample Target Semideviation Formula. *The target semideviation,* s_{Target}*, is:*

$$s_{Target} = \sqrt{\sum_{\text{for all } X_i \le B}^{n} \frac{(X_i - B)^2}{n-1}}, \qquad (6)$$

where B is the target and n is the total number of sample observations. We illustrate this in Example 3.

EXAMPLE 3

Calculating Target Downside Deviation

Consider the monthly returns on a portfolio as shown in Exhibit 11:

Exhibit 11: Monthly Portfolio Returns

Month	Return (%)
January	5
February	3
March	−1
April	−4
May	4
June	2
July	0
August	4
September	3
October	0

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1. Calculate the target downside deviation when the target return is 3 percent.

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Solution:

Month	Observation	Deviation from the 3% Target	Deviations below the Target	Squared Deviations below the Target
January	5	2	—	—
February	3	0	—	—
March	−1	−4	−4	16
April	−4	−7	−7	49
May	4	1	—	—
June	2	−1	−1	1
July	0	−3	−3	9
August	4	1	—	—
September	3	0	—	—
October	0	−3	−3	9
November	6	3	—	—
December	5	2	—	—
Sum				**84**

Target semideviation = $\sqrt{\frac{84}{(12-1)}}$ = 2.7634%.

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2. If the target return were 4 percent, would your answer be different from that for question 1? Without using calculations, explain how would it be different?

Hide Solution

Solution:

If the target return is higher, then the existing deviations would be larger and there would be several more values in the deviations and squared deviations below the target; so, the target semideviation would be larger.

How does the target downside deviation relate to the sample standard deviation? We illustrate the differences between the target downside deviation and the standard deviation in Example 4, using the data in Example 3.

EXAMPLE 4

Comparing the Target Downside Deviation with the Standard Deviation

1. Given the data in Example 3, calculate the sample standard deviation.

Hide Solution

Solution:

Month	Observation	Deviation from the Mean	Squared Deviation
January	5	2.75	7.5625
February	3	0.75	0.5625
March	−1	−3.25	10.5625
April	−4	−6.25	39.0625
May	4	1.75	3.0625
June	2	−0.25	0.0625
July	0	−2.25	5.0625
August	4	1.75	3.0625
September	3	0.75	0.5625
October	0	−2.25	5.0625
November	6	3.75	14.0625
December	5	2.75	7.5625
Sum	27		**96.2500**

The sample standard deviation is $\sqrt{\frac{96.2500}{11}}$ = 2.958%.

2. Given the data in Example 3, calculate the target downside deviation if the target is 2 percent.

Month	Observation	Deviation from the 2% Target	Deviations below the Target	Squared Deviations below the Target
January	5	3	—	—
February	3	1	—	—
March	−1	−3	−3	9
April	−4	−6	−6	36
May	4	2	—	—
June	2	0	—	—
July	0	−2	−2	4
August	4	2	—	—
September	3	1	—	—
October	0	−2	−2	4
November	6	4	—	—
December	5	3	—	—
Sum				**53**

The target semideviation with 2 percent target = $\sqrt{\frac{53}{11}}$ = 2.195 percent.

3. Compare the standard deviation, the target downside deviation if the target is 2 percent, and the target downside deviation if the target is 3 percent.

Hide Solution

Solution:

The standard deviation is based on the deviation from the mean, or 27% / 12 = 2.25%. The standard deviation includes all deviations from the mean, not just those below it. This results in a sample standard deviation of 2.958 percent.

Considering just the four observations below the 2 percent target, the target semideviation is 2.195 percent. It is less than the sample standard deviation since target semideviation captures only the downside risk (i.e., deviations below the target). Considering target semideviation with a 3 percent target, there are now five observations below 3 percent, so the target semideviation is higher, at 2.763 percent.

Coefficient of Variation

We noted earlier that the standard deviation is more easily interpreted than variance because standard deviation uses the same units of measurement as the observations. We may sometimes find it difficult to interpret what standard deviation means in terms of the relative degree of variability of different sets of data, however, either because the datasets have markedly different means or because the datasets have different units of measurement. In this section, we explain a measure of relative dispersion, the coefficient of variation that can be useful in such situations. **Relative dispersion** is the amount of dispersion relative to a reference value or benchmark.

The coefficient of variation is helpful in such situations as that just described (i.e., datasets with markedly different means or different units of measurement).

Coefficient of Variation Formula. *The* ***coefficient of variation*** *(CV) is the ratio of the standard deviation of a set of observations to their mean value:*

$$CV = \frac{s}{\overline{X}} \qquad 7$$

where s is the sample standard deviation and $\overline{X}$ *is the sample mean.*

When the observations are returns, for example, the CV measures the amount of risk (standard deviation) per unit of reward (mean return). An issue that may arise, especially when dealing with returns, is that if $\overline{X}$ is negative, the statistic is meaningless.

The CV may be stated as a multiple (e.g., two times) or as a percentage (e.g., 200 percent). Expressing the magnitude of variation among observations relative to their average size, the CV permits direct comparisons of dispersion across different datasets. Reflecting the correction for scale, the CV is a scale-free measure (i.e., , it has no units of measurement).

We illustrate the usefulness of CV for comparing datasets with markedly different standard deviations using two hypothetical samples of companies in Example 5.

EXAMPLE 5

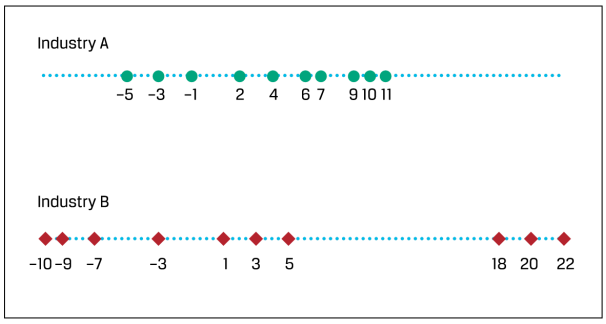
Coefficient of Variation of Returns on Assets

Exhibit 12: Returns on Assets for Two Industries

Company	Industry A	Industry B
1	−5	−10
2	−3	−9
3	−1	−7
4	2	−3
5	4	1
6	6	3
7	7	5
8	9	18
9	10	20
10	11	22

These data can be represented graphically as shown in Exhibit 13:

Exhibit 13: Returns on Assets Depicted Graphically



1. Calculate the average ROA for each industry.

Hide Solution

Solution:

The arithmetic mean ROA for both industries is the sum of the 10 observations, which in both cases is 40, divided by the 10 observations, or 40/10 = 4%.

2. Calculate the standard deviation of ROA for each industry.

Hide Solution

Solution:

Using Equation 5, the standard deviation for Industry A is 5.60 and 12.12 for Industry B.

3. Calculate the coefficient of variation (CV) of the ROA for each industry.

Hide Solution

Solution:

Using Equation 7, the CV for Industry A = 5.60/4 = 1.40 and the CV for Industry B = 12.12/4 = 3.03.

Though the two industries have the same arithmetic mean ROA, the dispersion is quite different—with Industry B's returns on assets being much more disperse than those of Industry A. The CV for these two industries reflects this, with Industry B having a larger CV. The interpretation is that the risk per unit of mean return is more than two times (2.16 = 3.03/1.40) greater for Industry B than Industry A.

Question Set

Consider the annual MSCI World Index total returns for a 10-year period, as shown in Exhibit 14:

Exhibit 14: MSCI World Index Total Returns

Year 4	9.57%	Year 9	16.54%
Year 5	−40.33%	Year 10	27.37%

1. For Years 6 through 10, the mean absolute deviation (MAD) of the MSCI World Index total returns is *closest* to:

A. 10.20 percent.
B. 12.74 percent.
C. 16.40 percent.

Hide Solution

Solution:

A is correct. The MAD is calculated as follows:

Step 1 Sum annual returns and divide by *n* to find the arithmetic mean $(\overline{X})$ of 16.40 percent.

Step 2 Calculate the absolute value of the difference between each year's return and the mean from Step 1. Sum the results and divide by *n* to find the MAD:

$$\text{MAD} = \frac{\sum_{i=1}^{n} |X_i - \overline{X}|}{n},$$

$$\text{MAD} = \frac{50.98\%}{5} = 10.20\%.$$

These calculations are shown in the following table:

	Step 1	Step 2
Year	**Return**	$\lvert X_i - \overline{X}\rvert$
Year 6	30.79%	14.39%
Year 7	12.34%	4.06%
Year 8	−5.02%	21.42%
Year 9	16.54%	0.14%
Year 10	27.37%	10.97%
Sum:	82.02%	50.98%
n:	5	5
$\overline{X}$:	16.40%	10.20%

Annual returns and summary statistics for three funds are listed as follows:

	Annual Returns (%)		
Year	**Fund ABC**	**Fund XYZ**	**Fund PQR**
Year 1	−20.0	−33.0	−14.0
Year 2	23.0	−12.0	−18.0
Year 3	−14.0	−12.0	6.0
Year 4	5.0	−8.0	−2.0
Year 5	−14.0	11.0	3.0
Mean	−4.0	−10.8	−5.0
Standard deviation	17.8	15.6	10.5

2. The fund with the highest mean absolute deviation (MAD) is Fund:

A. ABC.
B. XYZ.
C. PQR.

Hide Solution

Solution:

A is correct. The MAD of Fund ABC's returns is the highest among the three funds. Using Equation 3, the MAD for each fund is calculated as follows:

MAD for Fund ABC =

$$\frac{|-20-(-4)|+|23-(-4)|+|-14-(-4)|+|5-(-4)|+|-14-(-4)|}{5} = 14.4\%.$$

MAD for Fund XYZ =

$$\frac{|-33-(-10.8)|+|-12-(-10.8)|+|-12-(-10.8)|+|-8-(-10.8)|+|11-(-10.8)|}{5} = 9.8\%.$$

MAD for Fund PQR =

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Exhibit 15: Portfolio A and Portfolio B Statistics

	Portfolio A	Portfolio B
Average Return	3%	3%
Geometric Return	2.85%	?
Standard Deviation	4%	6%

The geometric mean return of Portfolio B is *most likely* to be:

A. less than 2.85 percent.
B. equal to 2.85 percent.
C. greater than 2.85 percent.

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Solution:

A is correct. The higher the dispersion of a distribution, the greater the difference between the arithmetic mean and the geometric mean.

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FA For exhibit 13 question 2, they need sample SD? or the Population, and how i know which one is asked

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VG This might be a dumb question but where are you getting the returns for Fund ABC, XYZ and PQR for Exhibit 14 Question 2. I can only see the MSCI World Index Total Returns

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FA For question 2 exhibit 14 world index returns. Can you just use the CV in which you dividing sd/mean? can you assume that the higher cv the greater deviation from mean?

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Exhibit 15. How can the Geometric mean return of a portfolio with lesser volatility of 4% be higher than that of a portfolio with higher volatility of 6% ?

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CS Example 4-3's answer gives an explanation for Mean Absolute Deviation, not standard deviation. The question should have read "Sample standard deviation" or "Mean Absolute Deviation" instead of Standard Deviation for the given solution.

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NA I don't understand Q4, Exhibit15

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MO In exhibit 14: is the sample mean calculated incorrectly? The arithmetic mean does not seem accurate. I am getting 9.718%

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